

Research paper

Catalytic pyrolysis of mixed plastic wastes from public places to liquid oil: A case of Kathmandu Valley, Nepal

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ABSTRACT

This study addresses the challenge of plastic waste disposal by exploring its potential as an alternative fuel source. It investigates the conversion of a plastic waste mixture (20 % LDPE, 40 % HDPE, and 40 % PET) into liquid fuel through catalytic pyrolysis. Using a custom-designed small-scale pyrolysis plant, the plastic samples were converted into liquid fuel with the aid of locally sourced bentonite powder as a catalyst, at temperatures ranging from 250 °C to 450 °C.

Results demonstrate that the oil yield increases with both temperature and use of bentonite powder as a catalyst. At 450 °C, 351±7.64 ml of liquid oil was produced with 10 g of catalyst, increasing to 355±4.17 ml with 15 g and 358±2.65 ml with 20 g, using 1 Kg of plastic waste sample. The retention time is reduced from 1.5 h without a catalyst to 30 min with it. A 20 % liquid oil and 80 % gasoline blend demonstrated comparable power, torque, and engine temperature with respect to standard gasoline, achieving a mileage of 19.64±1.55 Km/L, engine torque of 1.37 Nm and engine temperature of 110 °C at 3000 rpm, though higher blends show diminished performance.

Key limitations include a low cetane number (43.8), use of old motorcycles, and lack of advanced chemical analysis (GC–MS/FTIR). Despite this, the study bridges lab research to practical application via intermediate-scale experiments and real-engine tests with a low-cost local catalyst. While further refining and emissions testing are needed, it highlights pyrolysis oil's potential as a sustainable fuel in the future.

1. Introduction

Plastics have become indispensable in modern society due to their versatility, durability, and cost-effectiveness. However, this growing reliance has led to an exponential rise in their production and consumption. Over the past century, global plastic production has increased 20-fold, and projections suggest that annual output could exceed 500 million tons by 2050, consuming up to 40 % of the world's crude oil [1]. This surge in production has led to an unprecedented accumulation of plastic waste, creating a global environmental crisis. Each year, 4.8 to 12.7 million tons of plastic enter the oceans [2], forming massive oceanic gyres. The detection of microplastics in human blood [3] and breast milk [4], underscores the pervasive and insidious nature of plastic

pollution. Despite decades of recycling efforts, only 9 % of the 6300 million metric tons of plastic produced by 2015 [5] had been recycled, highlighting the urgent need for more effective waste management strategies.

The challenges of plastic waste management are particularly acute in rapidly urbanizing city such as Kathmandu Valley, Nepal. Daily It generates approximately 523.8 t of solid waste daily, 12 % of which is plastic [6]. Landfilling remains the dominant disposal method, with major sites including Okharpauwa and Teku [7]. The recent development of the Banchare Danda landfill site highlights the continued reliance on landfilling, despite its limitations. Unmanaged landfill sites can become breeding grounds for pathogens, exacerbating public health concerns and reducing the operational lifespan of these facilities.

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Unscientific dumping practices and inadequate waste segregation further accelerate the depletion of landfill capacity and increase environmental hazards. Policy interventions, such as 2015 declaration of Kathmandu as a polythene bag-free zone, have achieved limited success, as plastic bags remain widely used. This persistence reflects the socio-economic complexity of addressing plastic pollution in urban settings. While alternatives like glass, wood, bamboo, metals were proposed, their higher production costs significantly burdened of marginalized communities, limiting large-scale adoption. As urbanization continues and the availability of land for landfill disposal diminishes, urban cities like Kathmandu faces growing pressure to transition toward more sustainable and waste management solutions [8].

Pyrolysis has emerged as a promising technology for converting plastic waste into valuable energy products. This process involves the thermal degradation of plastics in an oxygen-free environment, transforms waste into solids, liquids, and gases, offering a sustainable alternative to traditional disposal methods. Operating within a temperature range of 300–500 °C [9], it can convert plastic waste into oil, providing a viable pathway for energy recovery. Catalysts such as ZSM-5, zeolite, and FCC catalysts enhance process efficiency, lowering thermal degradation temperatures, reducing activation energy, and optimizing conversion rates [10]. In atmospheric pressure experiments, inert gases like N₂, He, and Ar are used to remove oxygen from the system [11] while recent research has explored alternatives like H₂ and CO₂ [11]. These advancements make catalytic pyrolysis an attractive solution for addressing the global plastic waste crisis. Nevertheless, significant challenges remain. Pyrolysis is energy-intensive, which can reduce its net efficiency, and uncontrolled operations may emit toxic gases and particulate matter. Additionally, contaminants such as PVC can produce carcinogenic dioxins and furans, underscoring the need for strict process control and feedstock management.

Previous studies have demonstrated potential of pyrolysis for plastic waste conversion and energy recovery. For instance, a mixture of 45 % polyethylene (PE), 20 % polypropylene (PP), 20 % polystyrene (PS), and 15 % polyethylene terephthalate (PET) yielded 45 % pyrolysis oil by weight at 600–700 °C without a catalyst [12]. Similarly, a 70 % PE and 30 % PS mixture produced 93.7 % pyrolysis oil at 300–500 °C [13]. Catalytic pyrolysis using a 30 % PET and 70 % PS mixture with 20 % Al₂O₃ as a catalyst yielded 22.5 % oil at 250–500 °C [14]. Microwave-assisted pyrolysis of low-density polyethylene (LDPE) using ZSM-5 at 450 °C resulted in a 32.58 % oil yield [15]. While these studies highlight the potential of pyrolysis, they are largely confined to small-scale laboratory experiments, limiting their applicability to real-world scenarios [11].

Most existing research on catalytic pyrolysis predominantly focuses on small-scale experiments, typically using 5–200 g of plastic under idealized laboratory conditions [11]. Few studies explore larger-scale systems or test the resulting fuels in real engines. Instead, most research relies on simulation engines, leaving a significant knowledge gap regarding pyrolysis-derived fuels performance in actual machinery [16]. This discrepancy between laboratory-scale studies and practical applications hinders the scalability and commercialization of pyrolysis technology.

The present study addresses this gap by implementing a pyrolysis system that closely reflects the industrial process and utilize realistic mixed plastic wastes collected from public areas of Kathmandu Valley, rather than laboratory-grade samples. This approach, particularly novel and relevant in developing countries like Nepal, provides realistic understanding of product yields compared to studies using controlled, laboratory-grade samples. A small-scale pyrolysis plant capable of processing mixed plastic waste was constructed and optimized to replicate the industrial operating conditions, thus offering valuable insights into scalability of the process. The study focuses on catalytic pyrolysis using a raw Bentonite powder— a cost-effective, locally available catalyst—as an alternative to expensive activated catalysts.

Additionally, the physical properties of the produced oil—including density, viscosity, flash point, and cetane number—were analyzed and compared with conventional standard fuels. Furthermore, the oil was tested in a real motorbike engine, a common mode of transportation in Kathmandu, to evaluate its practical feasibility, efficiency and blending limitations. By bridging the gap between laboratory-scale research and practical application, this study provides new insights into the feasibility and scalability of pyrolysis as a sustainable waste-to-energy solution and emphasizes its potential as an alternative fuel for developing urban regions facing challenges with waste management and fuel security.

While this study provides a valuable foundation for full-scale exploration under certain conditions, it has several limitations. The relatively high PET content (40 %) in the mixed plastic waste reduced liquid yield due to PET's resistance to catalytic pyrolysis. The absence of an inert atmosphere and manual stirring led to gas leakage, lowering pyrolysis efficiency. The use of unactivated Bentonite catalyst may have limited catalytic activity, limiting overall conversion. Engine performance relied on torque measurements from a single old motorbike, introducing potential variability and limiting statistical rigor. Moreover, fuel characterization omitted advanced analyses like GC-MS, FTIR, calorific value, and emission testing, restricting insights into chemical composition and environmental impact of the produced fuel. Addressing these gaps in future studies will enhance the reliability, scalability and sustainability of plastic pyrolysis-derived fuels.

2. Materials and methods

2.1. Plastic sample collection and preparation

Commercial-grade Bentonite clay, in its powdered raw form, was procured from local vendors, chosen for its catalytic properties and accessibility. The primary feedstock—waste plastics—was sourced from Doko Recyclers Pvt. Ltd., a renowned entity specializing in the collection and recycling of plastic waste in the Kathmandu Valley, encompassing the districts of Kathmandu, Bhaktapur, and Lalitpur. Plastics were systematically collected on a weekly basis, to ensure representative sampling of waste, comprising high-density polyethylene (HDPE), low-density polyethylene (LDPE), polyethylene terephthalate (PET), and minor traces of acrylonitrile butadiene styrene (ABS). Post-collection, the plastic waste were thoroughly washed to remove surface contaminants and residues, then dried under direct sunlight for an entire day to eliminate residual moisture and optimize conditions for pyrolysis. After drying, the plastics were mechanically shredded into uniform fragments of approximately 4–5 cm². This size reduction was critical to ensure homogeneity and promote efficient thermal degradation and fuel conversion in the pyrolysis reactor. The systematic preparation process—from acquisition to shredding—laid the foundation for the subsequent experimental procedures, ensuring the integrity and reliability of the results.

2.2. Experimental set-up

The core of the experimental setup was a batch reactor crafted from mild steel into a cylindrical form, with a diameter of 400 mm, a height of 600 mm, and a maximum feed capacity of 10 kg. This cylindrical vessel was divided into two distinct sections: the lower segment, extending up to a height of 400 mm, as a storage for feedstock, while the upper segment was equipped with a Galvanized Iron (GI) conduit pipe to channel pyrolysis vapors the condenser. This section also housed a digital temperature probe and a manually operated stirrer extending to the base of the reactor ensuring thorough mixing of the feedstock. The lower and upper parts were bolted together using fifteen 12 mm bolts and nuts. A high-temperature-resistant silicone gel was applied at the joint of the two sections to minimize gas leakage, however complete sealing was not achieved due to screw-type arrangement at the stirrer. To reduce leakage, stirring was performed only after the reactor reached

peak temperature and limited to three rotations before being stopped.

The reactor was connected to a shell-and-tube condenser (150 mm diameter, and 1500 mm length) designed for efficient vapor condensation. Cold water was circulated through the 12 copper tubes (10.93 mm inner diameter) inside the condenser, while plastic vapors passed through the shell side. Twenty-eight baffles, spaced 52 mm apart, were installed to maximize heat exchange and condensation efficiency.

Heating was provided by a high-pressure Liquid Propane Gas (LPG) burner (Fig. 1), supplying the thermal energy required to initiate and sustain pyrolysis. Reaction time was monitored using a stopwatch, and volume of the liquid products was measured with a volumetric flask to ensure accurate yield quantification.

2.3. Seeding and operation of the reactor

Initially, the reactor was charged with 1 kg of mixed plastics, comprising 200 g of LDPE, 400 g of HDPE, and 400 g of PET. The experiments were conducted both with and without the presence of a catalyst to assess its impact on the pyrolysis process.

To maintain a uniform reaction temperature, the reactor was operated in a closed, airtight condition, which facilitated the gradual increase in pressure within the vessel. This increase in pressure, in turn, elevated the temperature inside the reactor. The reaction temperatures were regulated at three distinct levels: 250 °C, 350 °C, and 450 °C, to evaluate the influence of temperature on the yield of the resulting products.

Catalytic pyrolysis experiments were conducted at three different loadings: 10 g/Kg, 15 g/Kg and 20 g/Kg of plastic feedstock. Each experiment was conducted under identical conditions to the non-catalytic runs to enable direct comparison. Upon completion of each run, liquid oil was collected from the condenser, while the solid residue was retrieved from the bottom of the reactor after cooling. Residual wax deposited along the conduit walls was carefully scraped and collected before preparing the reactor for the next experiment. Liquid yield was measured using a volumetric flask, and the masses of carbon black residue, wax, and liquid oil were determined using a precision weighing balance. The total reaction time was measured with a stopwatch, defined as the interval from ignition until liquid dripping from the condenser ceased.

All measurements were recorded for both catalytic and non-catalytic experiments across the different temperature conditions and catalyst loadings. The choice of relatively low catalyst loadings was deliberate, aiming to minimize operational costs—an important consideration in the economically constrained context of Kathmandu.

2.4. Laboratory testing of the liquid oil

The characterization of the fuel properties of the catalytic pyrolysis oil was conducted at the Central Laboratory of Nepal Oil Corporation, employing a suite of standard analytical methods. The density of the pyrolysis oil was determined in accordance with ASTM D1298, which provides a method for measuring the density of petroleum products at a reference temperature of 15 °C. The kinematic viscosity, indicative of the oil's flow characteristics, was assessed at 40 °C using the ASTM D445 standard. The flash point, a measure of the oil's volatility and flammability, was evaluated using the IP 170 standard. Similarly, the pour point which indicates the lowest temperature at which the oil remains fluid was determined by ASTM D5949. The cetane number, a measure of the ignition quality of the fuel, was assessed using ASTM D613. The results obtained were compared with the standard values for diesel and gasoline using procedure outlined in the Indian Standard of Specification for Diesel and Gasoline.

2.5. Testing of fuel on gasoline engine

The pyrolysis oil derived from the catalytic conversion of plastic waste was blended with standard gasoline to form two distinct mixtures. The first mixture comprised 20 % pyrolysis oil and 80 % standard gasoline, while the second mixture consisted of 40 % pyrolysis oil and 60 % standard gasoline, as illustrated in Fig. 2a.

Engine performance tests were conducted using a 12-year-old Honda Shine motorcycle with an engine displacement of 124.7 mL and a body weight of 123 kg (Fig. 2b). The engine can deliver power output of 7.88 kWh at 7500 rpm and a torque of 10.3 Nm at 5500 rpm. The power transmission system of the motorcycle is equipped with gears, featuring a primary reduction value of 3.35, a gear ratio of 3.416 in 1st gear, and a final reduction ratio of 2.928 [17]. The use of an older motorcycle was a deliberate choice based on economic and practical considerations. The unrefined pyrolysis oil carried a risk of engine damage due to potential impurities, and testing on an older engine mitigated the cost of potential damage. Moreover, older motorcycles are common in developing regions and better reflect the real-world conditions under which such alternative fuels might be deployed. This study was therefore designed as a preliminary feasibility assessment, with future testing on newer engines planned once the fuel properties and operational parameters are better understood.

2.5.1. Measurement of the engine torque and power

Engine torque and power were evaluated as functions of engine shaft rotational speed (rpm), as both parameters vary dynamically with

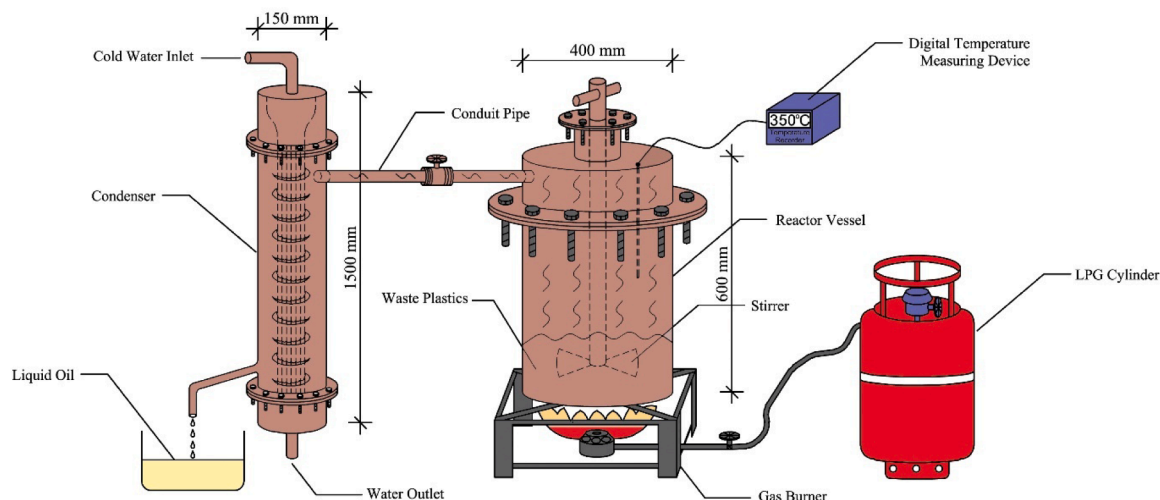


Fig. 1. Schematic diagram of the laboratory setup of pyrolysis reactor and condenser.

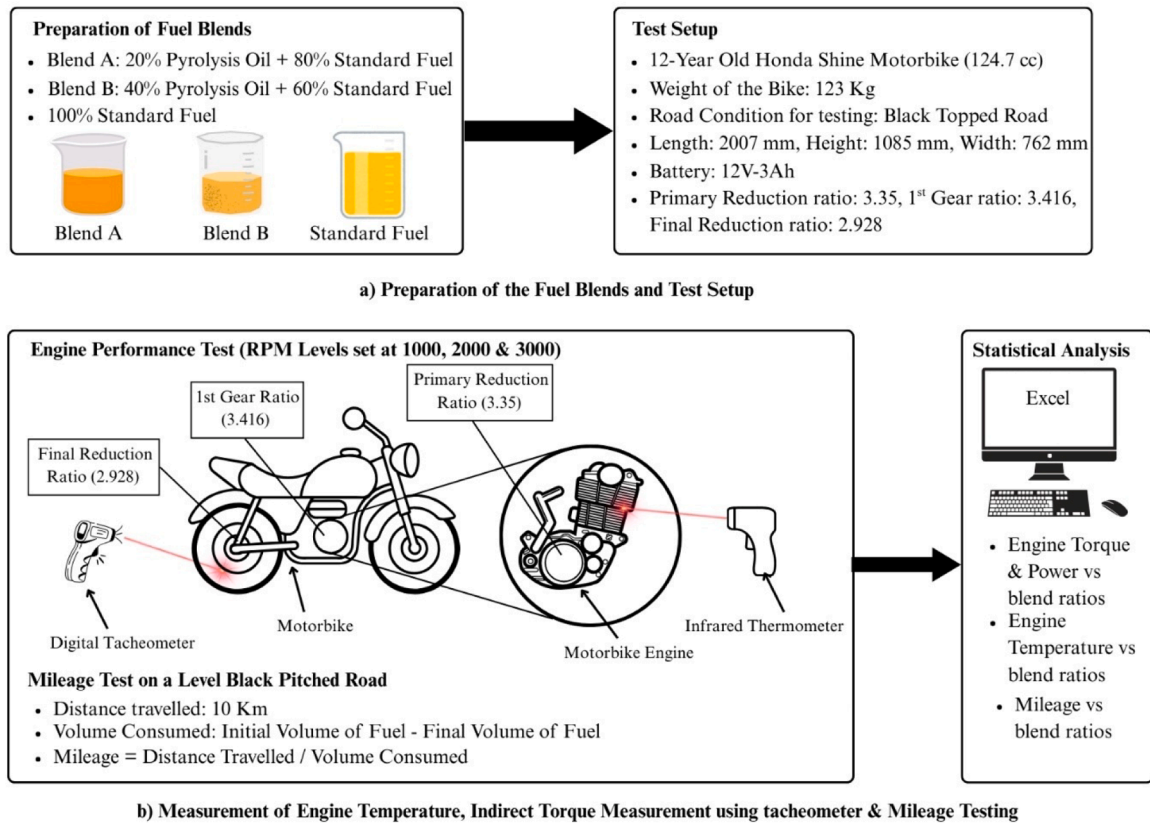


Fig. 2. Methodological Framework for the Engine Testing with Blended Fuel: (a) Preparation of Fuel Blends and Test Setup, (b) Measurement of Engine Temperature, Mileage, Torque and Statistical Analysis.

changes in rpm. The generated power and torque are transmitted to the rear wheel through the clutch assembly, gearbox, and chain-sprocket system. Each of these components is defined by its primary reduction, gear ratios, and final reduction ratio, which collectively govern the effective torque and power delivered to the wheel.

Due to practical challenges in directly measuring the engine shaft rpm, an indirect method was employed. The rotational speed of the rear wheel was measured using a digital tachometer, and this value was used to estimate engine torque and power. Because of the transmission system, the rpm at the rear wheel is lower than the engine shaft rpm, while torque and power are proportionally increased according to the primary reduction, gear ratio, and final reduction. The measured rear-wheel rpm was converted to angular velocity using Eqn (1). Angular acceleration was then determined from the change in angular velocity over time (Eqn (2)). The mass moment of inertia of the rear wheel was calculated using Eqn (3). Using these parameters, the torque produced at the wheel was determined from Eqn (4), and engine torque was calculated using Eqn (5) by accounting for transmission ratios. Finally, engine power was calculated from Eqn (6), while the engine rpm was derived using Eqn (7). All these equations are referenced from University Physics Book [18].

$$\text{Angular Velocity } (\omega) = \frac{\text{Rpm of the Wheel} \times 2\pi}{60} \quad (\text{rad / sec}) \quad (1)$$

$$\omega_f = \omega_i + \alpha \times t \quad (2)$$

Where, ω_f = Final angular velocity

ω_i = Initial angular velocity

α = Angular acceleration

t = time taken

$$I_{\text{wheel}} = I_{\text{rubbertire}} + I_{\text{OuterrimofWheel}} + 6 \times I_{\text{AlloySpokes}} \quad (3)$$

Where, I_{wheel} = Mass moment of inertia of a complete wheel (Kg m^2)

$I_{\text{rubber tire}}$ = Mass moment of inertia of outer rubber of wheel (Kg m^2); (mass of rubber tire $\times$ radius of wheel²)

$I_{\text{Outer rim of Wheel}}$ = Mass moment of inertia of outer circular rim of wheel (Kg m^2); (mass of alloy outer rim of wheel $\times$ radius of wheel²)

$I_{\text{Alloy Spokes}}$ = Mass moment of inertia of alloy spokes of wheel (Kg m^2); (mass of alloy spokes $\times$ Length of alloy spokes² / 3)

$$\text{Torque at wheel } (T_{\text{wheel}}) = I \times \alpha \quad (4)$$

Where, I = Mass moment of inertia of a complete wheel,
 α = angular acceleration

$$\text{Torque at Engine } (T_{\text{engine}}) = \frac{T_{\text{wheel}}}{(\text{Primary reduction} \times \text{gear ratio} \times \text{final reduction})} \quad (5)$$

$$\text{Power at Engine (P}_{\text{engine}}) = T_{\text{engine}} \times \text{Rpm}_{\text{engine}} \times 2\pi/60 \quad (6)$$

$$\text{Where, Rpm}_{\text{engine}} = \text{Rpm}_{\text{wheel}} \times \text{Primary reduction} \times \text{Gear ratio} \\ \times \text{Final Reduction} \quad (7)$$

In order to comprehensively assess engine's performance, the rotational speed of the engine was systematically set to predetermined values of 1000, 2000, and 3000 rpm. Subsequently, the corresponding torque and power outputs were calculated. Tests were conducted using two blended fuels—20 % pyrolysis oil with 80 % gasoline and 40 % pyrolysis oil with 60 % gasoline—as well as standard gasoline. This allowed a direct comparison of torque and power generation between the blended fuels and conventional gasoline.

2.5.2. Measurement of the bike mileage

Motorcycle mileage was evaluated using three fuel formulations: (i) 20 % pyrolysis oil + 80 % gasoline, (ii) 40 % pyrolysis oil + 60 % gasoline, and (iii) 100 % gasoline (control). For each test, the fuel tank was filled with a known volume of the designated fuel mixture. The motorcycle was then driven over a predetermined distance under consistent operating conditions, and fuel consumption was measured. At the end of each run, the remaining fuel volume was recorded, enabling calculation of the total fuel consumed. The distance travelled was accurately recorded using the motorcycle's speedometer, ensuring the reliability of the data. By dividing the distance travelled by the volume of fuel consumed, the mileage of the motorcycle for each fuel formulation was determined.

2.5.3. Measurement of the engine temperature

Three fuel formulations were evaluated for the same combination of oil mixture as in mileage test. Three fuel formulations, corresponding to the same oil–gasoline mixtures used in the mileage test, were assessed. For each formulation, the motorcycle engine was fueled and started, then allowed to idle for 10 min, a duration sufficient to stabilize the operating temperature. Engine surface temperatures were measured using a digital infrared thermometer, which provided precise, non-invasive readings. Measurements were taken at the end of the 10-minute idling period, and the procedure was repeated for all three fuel formulations, allowing for a comparative analysis of engine temperatures across the different fuel blends.

2.6. Statistical analysis

The liquid oil yield at varying temperature and catalyst to feed ratio along with the mileage test of the engine were expressed as mean $\pm$ standard deviation. The engine temperature, engine power and torque were expressed as value of the single test. Although single test data is less reliable, it serves as a baseline study before testing on modern engines. A single set of experimental data was collected for engine testing to mitigate the risk of potential damage caused by repeated testing. Statistical analyses were conducted in Microsoft Excel 365. A Pearson's correlation coefficient analysis was performed to evaluate the strength and direction of the linear relationship between the variables (catalyst ratio and temperature) and the oil yield. Corresponding p -values were calculated to determine the statistical significance of the observed relationships. Scatter plots were generated for visual representation, along with best fit linear equations and coefficient of determination (R^2) values to quantify the goodness of fit. Similarly, a Two-Way ANOVA (without interaction) was performed to assess the individual effects of temperature and catalyst-to-feed ratio on the oil yield from catalytic pyrolysis. Also, effect sizes were quantified using Eta squared (η^2) to determine the proportion of variance explained by each factor. Statistical significance was set at a p -value < 0.05 .

3. Results

3.1. Waste plastic characterization and pyrolysis reactor operation

A 50 kg plastic waste sample was characterized to determine its composition, which consisted of 20 % LDPE, 40 % HDPE and 40 % PET, with negligible ABS. This specific combination was subsequently used as the feedstock for the pyrolysis experiments.

The experimental outcomes revealed a positive correlation between temperature and liquid oil yield. Without catalyst, liquid oil production increased from 210 ± 18.03 ml at a temperature of 250°C to 291 ± 9.64 ml at 350°C and 338 ± 7.64 ml at 450°C . The addition of bentonite powder (10 g per kg of feed) enhanced yields further, producing 248 ± 2.89 ml at 250°C , 305 ± 5.0 ml at 350°C , and 351 ± 7.64 ml at 450°C . Increasing catalyst loading to 15 g and 20 g per kg of feed produced marginally higher yields, reaching a maximum of 358 ± 2.65 ml at 450°C (Fig. 3).

In addition to increasing yield, the catalyst nearly halved the overall reaction time for complete pyrolysis (Fig. 4). However, no significant reduction in reaction time was observed when increasing catalyst dosage beyond 10 g per kg of feed. These findings highlight the pivotal role of temperature and catalytic intervention in enhancing the liquid oil yield and reaction time during the pyrolysis process. Table 1 presents the distribution of pyrolysis products (by weight %), with and without the use of catalyst.

3.3. Results of the statistical analysis

Pearson correlation analysis confirmed that both temperature and catalyst loading significantly influenced liquid oil yield. The correlation between catalyst loading (0, 10, 15, and 20 g per kg of feed) and liquid oil volume at constant temperatures of 250°C , 350°C , and 450°C (Fig. 5a-c) exhibited strong positive linear relationships, with R^2 values ranging from 0.88 to 0.99. The corresponding p -values (< 0.05) indicated statistical significance, confirming that increasing catalyst dosage enhances liquid oil production. However, at 250°C , the p -value exceeded 0.05, suggesting that lower temperatures limit catalytic activity and result in reduced oil yield even at higher catalyst loadings.

Similarly, the effect of temperature on oil yield at fixed catalyst ratios (Fig. 5d-f) showed consistently higher R^2 values and statistically significant p -values (< 0.05), indicating that higher pyrolysis temperature improves the yield of liquid oil regardless of catalyst amount.

These findings statistically validate that both the temperature and catalyst ratio are critical variables influencing the efficiency of pyrolysis. The observed linear trend and significant correlations support the hypothesis that optimized thermal and catalytic conditions can maximize liquid oil production from plastic waste.

Result from two-way ANOVA depicts that temperature had a much stronger effect on liquid oil yield compared to catalyst-to-feed ratio. The analysis showed higher F-Value and extremely low p -value for the temperature, indicating a statistically highly significant effect. In contrast, the catalyst-to-feed ration exhibited a relatively weaker influence on oil yield. Furthermore, the higher Eta Squared (η^2) for temperature confirmed temperature as a dominant factor in influencing the pyrolysis oil yield. This dominance can be attributed to the relatively low catalyst loadings used in the present study, which may have limited the observable effect of catalyst ratio on oil production. has contributed to showing significant effect of temperature on oil yield in our study. The detailed ANOVA results and η^2 values are summarized in Table 2.

3.4. Laboratory testing of the liquid oil

The physical properties of pyrolysis-derived liquid oil—density, kinematic viscosity, cetane number, and flash point—were determined. The density of the oil at 15°C was measured to be 816.6 kg/m^3 , which falls within the range specified by Indian standards for both standard

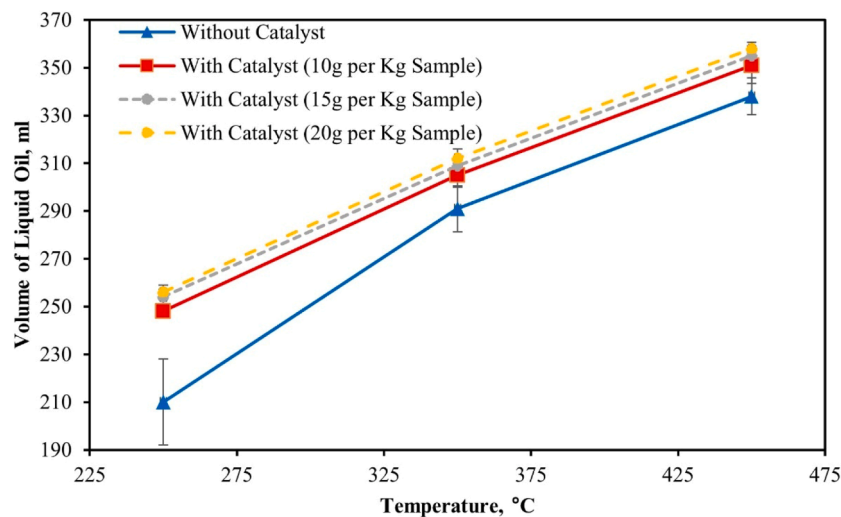


Fig. 3. Variation of the yield of the liquid oil with the temperature, with and without the use of the catalyst.

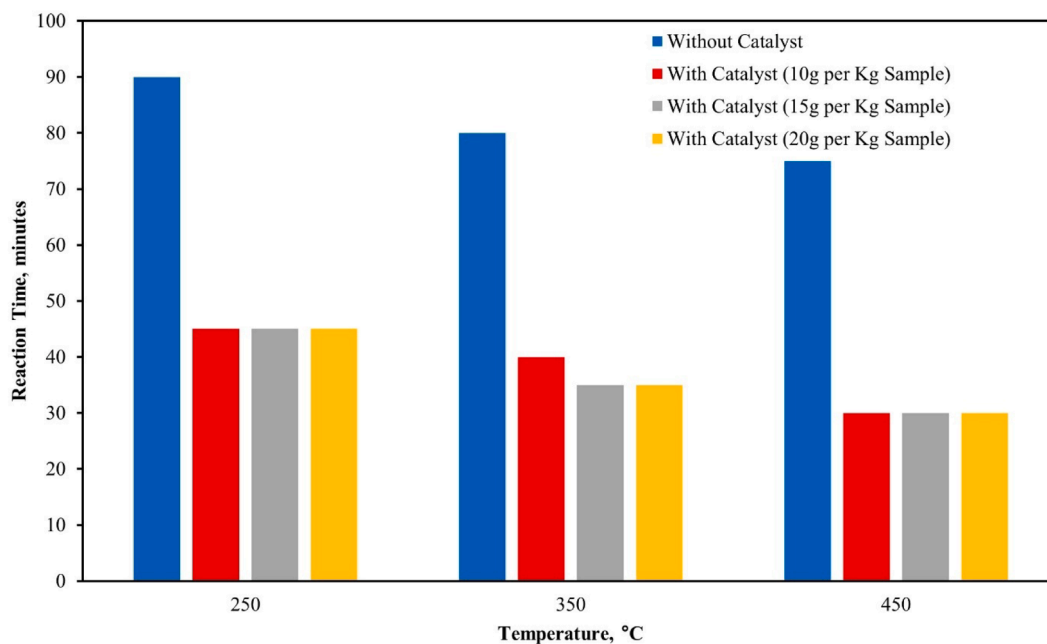


Fig. 4. Comparison of the reaction time of the pyrolysis at different temperature, with and without the use of the catalyst.

diesel and gasoline (Fig. 6a). The kinematic viscosity at 40 °C was 1.95 cSt, as shown in Fig. 6b, aligning with the viscosity range of conventional fuels. The flash point of the pyrolysis oil was measured at 53 °C (Fig. 6c), also within acceptable range for standard fuels. However, the cetane number was 43.8 (Fig. 6d), which is lower than that of standard diesel and gasoline, indicating a relatively inferior quality. Despite this limitation, the density, viscosity, and flash point measurements indicate that the pyrolysis oil could be a viable alternative for certain fuel applications, potentially after blending or upgrading to improve its ignition characteristics.

3.5. Analysis of the performance of the gasoline engine

Engine performance tests were conducted using various blends of pyrolysis-derived oil and standard gasoline to evaluate performance on a conventional gasoline engine. The primary objective was to compare torque and power output across different engine speeds (1000–3000 rpm) and blend ratios. The 20 % pyrolysis oil + 80 % standard gasoline

blend produced a torque of 0.78 Nm and a power of 0.08 kW at 1000 rpm, which increased to 1.19 Nm/0.25 kW at 2000 rpm and 1.89 Nm/0.59 kW at 3000 rpm (Fig. 7a,b). In contrast, 40 % pyrolysis oil + 60 % standard gasoline blend demonstrated reduced performance at all speeds. Torque and power outputs were 0.50 Nm / 0.05 kW at 1000 rpm, rising to 0.81 Nm / 0.17 kW at 2000 rpm and 1.37 Nm / 0.43 kW at 3000 rpm (Fig. 7a,b).

For benchmarking purpose, the engine was also tested standard gasoline which yielded highest performance. At 1000 rpm, torque and power were 1.01 Nm / 0.11 kW, increasing to 1.68 Nm / 0.35 kW at 2000 rpm and 2.33 Nm / 0.73 kW at 3000 rpm (Fig. 7a,b). These results suggest that the blended mixture of 20 % liquid oil with 80 % gasoline provides engine performance relatively close to conventional fuel, whereas higher blend ratios result in noticeable performance degradation. This finding indicates that moderate blending may be a feasible strategy for utilizing pyrolysis oil as an alternative fuel without significantly compromising engine efficiency.

Table 1

Product distribution of pyrolysis of plastic mixture with and without the use of catalyst.

SN	Feedstock	Operating Conditions	Quantity (in % by weight)		
			Liquid Oil	Carbon Black Residue	Wax
1	(40 % PET + 40 % HDPE + 20 % LDPE) without use of catalyst	T = 250 °C Time = 1.5 h	17.14	56.6	8.35
		T = 350 °C Time = 1.5 h	23.75	54.3	7.85
		T = 450 °C Time = 1.5 h	27.61	48.1	7.23
		T = 250 °C Time = 45 min	20.26	49.1	8.03
2	(40 % PET + 40 % HDPE + 20 % LDPE) 10 g of catalyst per Kg of Plastic waste	T = 350 °C Time = 40 min	24.89	45.0	7.65
		T = 450 °C Time = 30 min	28.7	44.7	6.53
		T = 250 °C Time = 45 min	20.70	49.10	8.43
		T = 350 °C Time = 35 min	25.19	45.10	7.65
3	(40 % PET + 40 % HDPE + 20 % LDPE) 15 g of catalyst per Kg of Plastic waste	T = 450 °C Time = 30 min	28.94	44.0	6.56
		T = 250 °C Time = 45 min	20.89	48.90	8.20
		T = 350 °C Time = 35 min	25.46	44.90	7.18
		T = 450 °C Time = 30 min	29.21	43.30	7.23
4	(40 % PET + 40 % HDPE + 20 % LDPE) 20 g of catalyst per Kg of Plastic waste	T = 250 °C Time = 45 min	20.89	48.90	8.20
		T = 350 °C Time = 35 min	25.46	44.90	7.18
		T = 450 °C Time = 30 min	29.21	43.30	7.23
		T = 250 °C Time = 45 min	20.89	48.90	8.20

3.6. Thermal analysis of engine operation

Engine temperature measurements were recorded for each fuel blend across speeds of 1000–3000 rpm to assess the thermal behavior of the engine under varying operating conditions. For the 20 % pyrolysis oil + 80 % gasoline blend, the engine temperature stabilized at 80 °C at 1000 rpm, rising to 93 °C at 2000 rpm and 103 °C at 3000 rpm (Fig. 8a). The 40 % pyrolysis oil + 60 % gasoline blend exhibited a consistently higher thermal profile. At 1000 rpm, the temperature reached 84 °C, increasing to 99 °C at 2000 rpm and peaking at 110 °C at 3000 rpm (Fig. 8a). In comparison, standard gasoline yielded the lowest temperature readings across all speeds, with 75 °C at 1000 rpm, 87 °C at 2000 rpm, and 93 °C at 3000 rpm (Fig. 8a). These findings indicate that increasing the proportion of pyrolysis oil in the blend leads to higher engine operating temperatures, likely due to differences in combustion characteristics between pyrolysis oil and conventional gasoline.

3.7. Mileage efficiency analysis

Mileage tests were performed to determine the effect of pyrolysis oil blending on vehicle fuel economy. For the 20 % pyrolysis oil + 80 % gasoline blend, the mileage was 19.64 ± 1.55 km/L (Fig. 8b). Increasing the proportion of pyrolysis oil to 40 % further reduced mileage to 16.46 ± 1.03 km/L (Fig. 8b), reflecting the challenges posed by higher pyrolysis oil content—likely due to its lower energy density and different combustion characteristics compared to gasoline. By contrast, pure gasoline yielded the highest mileage at 34.69 ± 2.35 km/L, serving as the benchmark for comparison (Fig. 8b). This result underscores the superior fuel efficiency of conventional gasoline under the same operating conditions.

4. Discussion

4.1. Impact of PET on oil yield and environment

Oil yield during pyrolysis is highly dependent on the type and composition of plastics used. In a study on pyrolysis of PET, PS, PE and PP in a pilot reactor operated at 450 °C with 45-minutes retention time, PET produced significantly lower liquid oil yields compared to HDPE and LDPE plastics [19]. This is attributed to the aromatic rings in PET, which imparts structural stability and make it less prone to thermal degradation into liquid products [20]. The phenylene (-pH-) rings in PET form conjugated π -systems, and breaking these bonds requires substantially more energy than cleaving aliphatic chains. Consequently, PET favors the formation of stable char and gaseous products rather than liquid hydrocarbons. In contrast, the simpler aliphatic structure of HDPE and LDPE promotes their breakdown into condensable liquid fractions [21]. Additionally, the presence of higher oxygen content in PET leads to the formation of more non-condensable gases and solid residues [22].

Mixed-plastics pyrolysis also contributed to the low oil yield. Odejobi et al. reported a ~30 % reduction in oil yield when co-pyrolyzing LDPE with PET [23]. PET-derived radicals scavenge the hydrogen-rich radicals from LDPE/HDPE, suppressing chain growth into liquid hydrocarbons and diverting carbon into char. The relatively low oil yield in our study is likely due to presence of 40 % of PET in the feedstock.

From an environmental perspective, PET pyrolysis produces polycyclic aromatic hydrocarbons (PAHs) and biphenyl derivatives, many of which are carcinogenic [24]. These compounds, if released into the atmosphere can form organic aerosols posing inhalation risks and contribute to smog formation. During our experiments, a foul odor and eye irritation were noted, suggesting the release of toxic gases. Although PET contains no chlorine, mixed-feed reactors can facilitate the formation of dioxins, furans, and polychlorinated biphenyls, which are highly persistent and bio-accumulative if not properly controlled [25]. The large amount of solid residue formed from PET pyrolysis contains adsorbed PAHs and heavy metals [24], which if landfilled without treatment can risk the soil and groundwater contamination.

Despite these challenges, the carbon-rich solid residue holds considerable industrial value. It can be utilized as a reinforcing filler in rubber products such as tires, conveyor belts and footwear, improving enhances strength and abrasion resistance [26]. Its high surface area and electrical conductivity also make it promising for energy storage applications, including supercapacitors and lithium-ion batteries [27].

4.2. Effect of temperature, inert atmosphere and gas leakage on oil yield

The liquid yield is also affected by the reaction temperature. For instance, a study on HDPE pyrolysis at 400 °C, 425 °C, and 450 °C reported a progressive increase in liquid yield, with the maximum obtained at 450 °C [28]. This trend is attributed to the enhanced breakdown of polymer chains into smaller hydrocarbon molecules at elevated temperatures. Our results follow a similar trend: increasing the temperature from 250 °C to 450 °C significantly improved oil yield by providing sufficient the thermal energy required for polymer breakdown. However, the maximum temperature in our study is limited to 450 °C, as Shah et al. (2023) reported that beyond plastic threshold temperature (~500 °C), oil yield significantly decreases as more non-condensable gas is formed [29].

The use of a stirrer for mixing during pyrolysis likely contributed to gas leakage from the reactor, allowing vapors that should have been condensed to escape. The International Energy Agency has emphasized that preventing gas leakage is critical to maximizing liquid oil production [30]. Pyrolysis must be conducted under strictly anaerobic conditions; to maintain this environment, inert gases such as nitrogen, helium, or argon are commonly introduced. These gases displace oxygen, promote hydrocracking, reduce coke formation, and enhance catalyst efficiency [11]. In our system, gas leakage during stirring likely allowed air

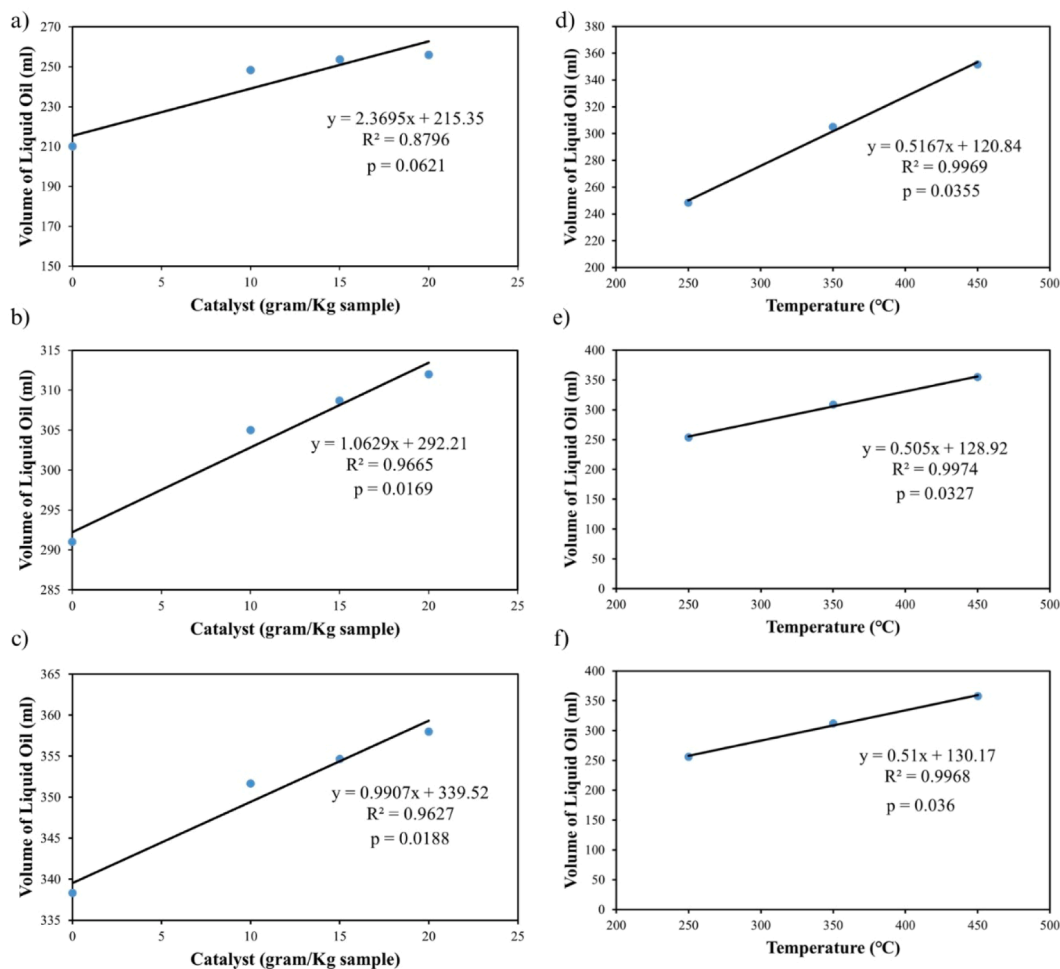


Fig. 5. Scatter Plot with R^2 and associated p-value a) Plot of volume of liquid oil at varying catalyst ratio at 250 °C b) Plot of volume of liquid oil at varying catalyst ratio at 350 °C c) Plot of volume of liquid oil at varying catalyst ratio at 450 °C d) Plot of volume of liquid oil at varying temperature at 10 g catalyst per kg sample e) Plot of volume of liquid oil at varying temperature at 15 g catalyst per kg sample f) Plot of volume of liquid oil at varying temperature at 20 g catalyst per kg sample.

Table 2

Two-way ANOVA results' summary (No Interactions).

Sources of Variation	Sum of Squares	Degree of Freedom	Mean Squares	F-value	p-value	Eta Squared (η^2)
Temperature	23,779.76	2	11,889.88	215.31	0.000003	0.927
Catalyst	1543.76	3	514.58	9.32	0.011238	0.060
Residual/Error	331.32	6	55.22			

ingress, disrupting the anaerobic environment and lowering process efficiency. The foul odor observed during operation further supports the occurrence of leakage. The combination of atmospheric contamination and vapor loss likely contributed to the reduced liquid oil yield observed in our study.

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reduced liquid oil yield observed in our study.

4.3. Catalyst efficiencies and its economic feasibility

The thermal degradation of plastics is energy-intensive; however, the use of catalysts can lower the activation energy, reducing the energy required for thermal cracking. Bentonite powder, a natural clay mineral, has been widely used as a catalyst in the plastic pyrolysis. Its acidic sites promotes cracking reactions, reducing reaction time and enhancing liquid oil yield [11]. The catalytic efficiency of bentonite depends on its surface area and pore structure. For instance, a study reported that bentonite possesses a BET surface area of 47 m²/g and a total pore volume of 0.13 cm³/g, with an average pore size of 10.9 nm, indicating a highly mesoporous structure [16]. These characteristics can limit the accessibility of large polymer molecules to the active sites, potentially explaining why variations in the catalyst-to-feed ratio have negligible effects on the pyrolysis process.

Catalysts are reported to have minimal impact on PET plastic due to

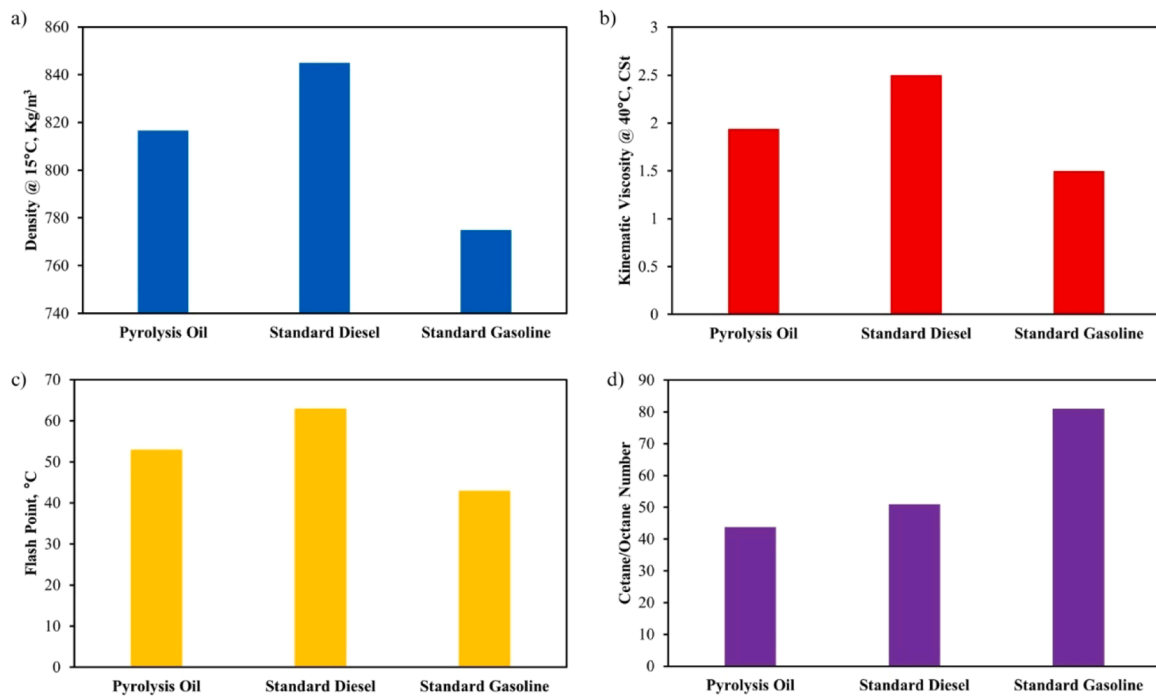


Fig. 6. Comparison of the density and kinematic viscosity of the pyrolysis oil with standard diesel and gasoline: a) Comparison of the density and b) Comparison of the kinematic viscosity c) Comparison of the Flash Point and d) Comparison of the Cetane Number.

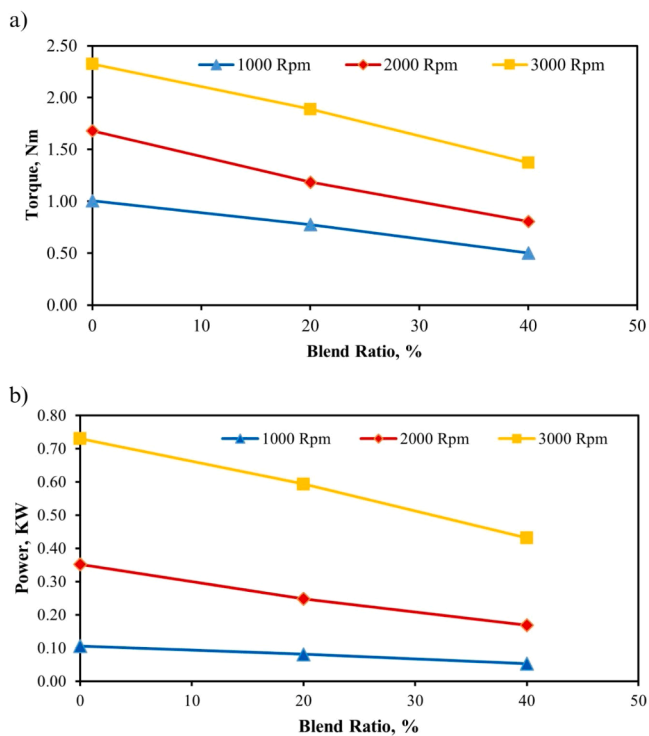


Fig. 7. Analysis of engine torque and power at different rpm and different blend ratios: a) Torque produced at different blend ratios and at different rpm values and b) Power produced at different blend ratios and at different rpm values.

its thermally stable aromatic ring structure [31]. Consistent with this observation, our results showed little difference in PET degradation regardless of catalyst use. In our study, bentonite reduced the pyrolysis reaction time from 90 min (without catalyst) to approximately 30 min.

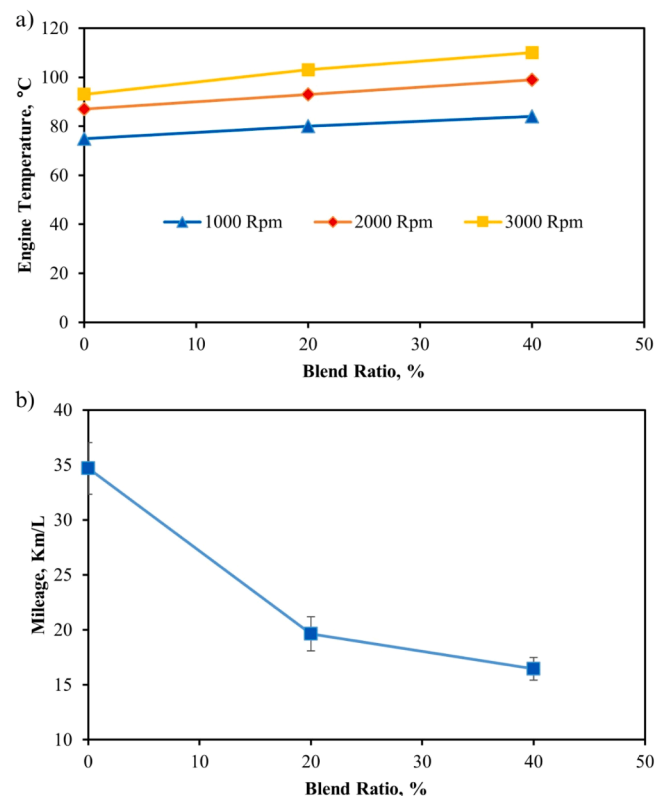


Fig. 8. Analysis of Thermal Performance and Mileage of the Motor Engine: a) Thermal behavior of the engine at different rpm values and at different blend ratios and b) Mileage of the Motor Bike at different blend ratios.

However, increasing the catalyst loading from 10 g to 15 g and 20 g per kg of plastic resulted in negligible changes in reaction time, suggesting a saturation point in catalytic activity. This observation suggests a

saturation point in catalytic activity, possibly due to limited catalyst surface area, the mesoporous structure restricting polymer accessibility and resistance of PET within the mixed plastic waste.

Raw bentonite powder was selected as a catalyst due to its cost-effectiveness and local availability. Although commercial activated catalysts—with higher surface areas and stronger Lewis acidity—could further enhance pyrolysis performance, they are not readily available in the Kathmandu Valley and are expensive to procure internationally. Additionally, many activated catalysts require precise dosing and controlled calcination, and those impregnated with transition metals or treated with strong acids pose inhalation, dermal, and environmental hazards. Their use therefore demands skilled technicians, further increasing operational costs. Despite these limitations, the use of bentonite catalyst in our study significantly reduced reaction time, resulting in substantial energy savings. Our findings confirm that the catalyzed process consumes much less energy than the non-catalyzed process, thereby improving both process efficiency and economic viability.

4.4. Physical properties of the liquid oil and its application on a motorbike engine

The pyrolysis oil exhibited physical properties within the range of standard diesel and gasoline for density, kinematic viscosity, and flash point, though the cetane number was below standard value. Most of the studies showed that the property of liquid oil is comparable to the standard diesel. Budsareechai et al. (2019) reported similar findings in catalytic pyrolysis, with the density, kinematic viscosity and flash point be 0.901 g/cm³, 2.0 cP and 57 °C for LDPE plastics and 0.91 g/cm³, 2.0 cP and 59 °C for HDPE plastics [16]. These values are in close range of standard diesel. The lower cetane number in pyrolysis oils stems from presence of aromatic compounds and oxygenated species, especially PET are used as feedstock [32]. The aromatic structure and higher oxygen content in PET contribute to a lower cetane number, reducing combustion efficiency in diesel engines.

Although the physical properties suggesting diesel engine compatibility, a gasoline engine (motorbike) was used for performance testing due to economic considerations, as diesel engine generally involves higher acquisition and operational costs for experimental purposes. This study was also limited by the unavailability of laboratory facilities for determining calorific value, gas chromatography-mass spectroscopy analysis, oxygen/sulfur content, acidity and stability parameters, which would have provided comprehensive quality assessment of the liquid oil.

The low cetane number of the pyrolysis-derived liquid oil indicates that further refining is required before commercial use. To evaluate its immediate applicability, blended mixtures of liquid oil and standard gasoline were tested on a bike engine under real-world conditions. Previous studies reported optimal performance at 30–50 % blend ratios in laboratory-controlled engine tests [33], but such studies relied on simulation models rather than practical operation. Our results showed a decrease in engine torque with higher pyrolysis oil content, consistent with earlier findings on direct-injection diesel engines [34]. The reduction is attributed to poor ignition quality, incomplete combustion, and poor atomization of pyrolysis oil [35]. In our case, high viscosity and aromatic hydrocarbon content from PET further impaired combustion, lowering torque and power.

The maximum rpm achieved was 3000, below the 7000-rpm capacity of newer models, likely due to the 12-year-old engine's reduced efficiency. Nevertheless, the aim was to compare power and torque between conventional fuel and pyrolysis oil blends, not to assess engine durability. Older engines still provide valuable baseline data for future tests on modern engines. Furthermore, unlike commercial gasoline with antioxidants, anti-knock agents, and friction modifiers, the unrefined pyrolysis oil lacked such additives, contributing to the observed performance losses.

Elevated viscosity in plastic pyrolysis oil negatively affects fuel

lubricity and engine performance. High viscosity leads to poor fuel atomization during injection, producing larger droplets that mix inefficiently with air and cause incomplete combustion [36]. This results in higher heat release, elevated engine temperatures, and increased residual heat. In our study, engine temperature rose with higher liquid oil ratios, likely due to both viscosity effects and impurities in the fuel mixture [37]. Post-test inspection revealed a damaged spark plug and a gasoline filter clogged with impurities, indicating complete combustion demand on the ignition system and inadequate filtration. These findings suggest that the liquid oil requires further refinement before commercial application.

A reduction in motorcycle mileage was also observed when using blended fuels, which can be attributed to the lower calorific value of pyrolysis oil compared to standard gasoline [38]. Lower energy content necessitates greater fuel consumption to achieve the same power output, thereby reducing efficiency. Impurities likely further diminished the effective energy content. According to the Bureau of Indian Standards IS 2796:2017, gasoline has a calorific value of 10,500–11,500 kcal/kg [39], whereas pyrolysis oil ranges from 15 to 36 MJ/kg [40], supporting the observed decrease in mileage. Calorific value measurements for our pyrolysis oil could not be performed due to the lack of appropriate equipment in the region. Similarly, emission testing was not conducted because Nepal does not yet have regulatory standards or laboratory facilities for two-wheeler emission analysis, though standards exist for four-wheeled vehicles.

4.5. Socio-Economic challenges and policy barriers in the context of Nepal

Despite its potential address Nepal's growing plastic waste problem and provide alternative fuel sources, the commercial application of plastic pyrolysis faces several socio-economic and policy-related challenges. High initial investment cost make the technology economically unfeasible without government subsidies or public-private partnerships. Moreover, the lack of technical manpower and limited public awareness hinder its social acceptance.

Nepal's Solid Waste Management Act (2011) provides a legal framework for emphasizing waste reduction, reuse, recycling and environmentally sound disposals, and supports waste-to-energy initiatives. However, it does not specifically address pyrolysis. The Solid Waste Management Rule (2013) encourages private sector participation in waste processing, which could facilitate the establishment of pyrolysis plants. Conversely, the Nepal Petroleum Act (1983) defines "petroleum" broadly to include synthetic fuels, meaning plastic-derived fuels would fall under its scope. The Act requires formal agreements with the government for production, treatment, transport, storage, and sale of petroleum products. Additionally, the Nepal Oil Corporation holds a monopoly over fuel import and distribution, creating significant market entry barriers for private entities seeking to sell alternative fuels such as those produced through pyrolysis.

Nevertheless, pyrolysis could provide an important alternative to conventional waste management in Kathmandu Valley where landfilling remains the dominant practice and poses significant environmental challenges. Pyrolysis offers a more sustainable and circular approach by thermochemically decomposing plastic waste into valuable products such as liquid oil and carbon black, without requiring extensive pre-sorting of the feedstock. As pyrolysis remains an unexplored technology in the region, this study delivers critical insights into its feasibility and establishes a framework for process optimization, offering a pathway for future development of sustainable waste-to-energy solutions in Nepal.

4.6. Energy efficiencies and industrial scale process improvements

The catalytic pyrolysis process in this study reduced reaction time from 90 to 30 min with bentonite catalyst, improving energy efficiency. Yet, industrial applications remain energy intensive, as sustaining

350–450 °C requires significant fuel. To address this, non-condensable syngas produced during pyrolysis can be recycled as heating fuel, reducing reliance on LPG or electricity by up to 30 % [9,10]. Similarly, heat recovery systems, such as condensers with heat exchangers and flue gas recuperators, capture thermal energy from vapors and exhaust gases that would otherwise be wasted [41].

Reactor design also influences efficiency. While batch systems are suitable for laboratory studies, continuous-feed reactors minimize heat loss, improve throughput, and are more practical for industrial scale-up [11]. The potential for catalyst regeneration further reduces operational costs by extending catalyst life and maintaining conversion activity [10]. In addition, innovative approaches like microwave-assisted [42] and solar-assisted pyrolysis are being explored, coupling renewable or direct heating with conventional energy sources. Although still at an experimental stage, these methods highlight pathways for reducing carbon intensity.

4.7. Environmental hazards and mitigations

While pyrolysis provides an alternative to landfilling, it can also generate environmental hazards if not properly managed. The degradation of PET and PVC may release PAHs, VOCs, dioxins, and furans, which are toxic and potentially carcinogenic [24]. Observations of odor and irritation during experiments suggest that such emissions occurred in small amounts. Furthermore, the carbon black residues can adsorb harmful organics and metals, posing risks of soil or water contamination if disposed of untreated.

To mitigate these risks, pyrolysis plants should incorporate gas cleaning technologies, including alkaline scrubbers, activated carbon filters, catalytic oxidation units, and multistage condensation systems. These reduce VOCs and acid gases while improving oil recovery. Residues should undergo toxicity and leachability tests to confirm their suitability for reuse or be stabilized if unsafe.

Equally important is systematic emissions monitoring, which was a limitation in this study. Future work should measure CO, CO₂, CH₄, NO_x, SO_x, particulate matter, and unburned hydrocarbons during pyrolysis, as well as emissions from engines using oil blends. In the absence of national standards, Euro VI norms can provide useful benchmarks. By addressing emissions and residues, pyrolysis can progress toward safer and more sustainable deployment.

4.8. Refining pyrolysis oil and comprehensive environmental assessments

The pyrolysis oil produced in this study showed acceptable density and viscosity, but its low cetane number (43.8) and impurities restrict direct use. Engine tests also indicated spark plug fouling and reduced efficiency at higher blending ratios. These findings highlight the need for refining before wider application.

Fractional distillation can separate lighter fractions with improved combustion quality, while hydrotreatment removes oxygenates and aromatics [12], enhancing stability and cetane value. Blending with conventional fuels or adding cetane improvers offers further improvement. In addition, advanced catalysts such as zeolites (HZSM-5, FCC) have shown superior ability to crack heavy hydrocarbons into lighter [14], more stable fractions compared to bentonite. Although more costly, their efficiency and regenerability make them suitable for industrial operations.

Equally important are comprehensive environmental assessments to ensure sustainability. These include chemical analysis (GC–MS, FTIR) to detect harmful compounds, toxicity testing of residues to confirm safe reuse or disposal, and calorific value with energy balance analysis to verify net energy gains. Broader evaluations such as Life Cycle Assessment (LCA) are also critical to compare pyrolysis with landfilling and incineration. By integrating refining steps with thorough assessments, catalytic pyrolysis can advance from laboratory success to a viable industrial solution.

5. Conclusions

This study demonstrates that catalytic pyrolysis process significantly reduces the reaction time and increases liquid oil yield compared to non-catalytic pyrolysis. Higher temperatures further enhanced oil production, while maintaining anaerobic conditions and minimizing gas leakage proved crucial for efficiency. However, variations in catalyst-to-feed ratio had negligible impact, and the presence of PET in the feed-stock reduced liquid yield. The physical properties of produced oil—density, viscosity and flash point—indicate potential for commercial application, but the low cetane number highlights the need for further refining processes to improve ignition quality. Preliminary tests involving the application of this liquid oil in a motorbike engine revealed promising performance, though additional post-treatment, such as hydrogenation or distillation, is required before large-scale commercial deployment.

Despite these limitations, this study provides a strong foundation for sustainable plastic management. Future work should focus on optimizing yields using activated catalysts, controlled inert atmospheres, and leak-proof stirring. Further work should also include calorific value assessment, GC–MS or FTIR analysis, emission testing, and post-treatment of pyrolysis oil prior to engine performance evaluation.

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Consent to participate

Not Applicable

Consent to publish

Not Applicable

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Declaration of generative AI in scientific writing

The authors have not used Generative AI tools for content generation while preparing this scientific manuscript. However, tools such as ChatGPT and Grammarly were used solely to enhance the grammatical clarity and readability of the sentences.

CRediT authorship contribution statement

Anish Pandey: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Tek Narayan Bhattarai:** Writing – review & editing, Visualization, Validation. **Narayan Prasad Adhikari:** Writing – review & editing, Validation, Methodology. **Shree Raj Shakya:** Writing – review & editing, Validation, Methodology. **Shukra Raj Paudel:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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